MATHEMATICS ALIVE – Geometry in Real Life	
	
Symmetry in Plants	Symmetry in architecture

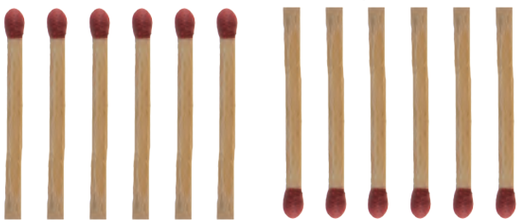
4.2 Symmetry through transformations

We learnt the types of symmetry. Now we are going to learn symmetry of figures through transformations.

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Transformation describe how geometric figures of the same shape are related to one another. Figures or shapes in a plane can be translated, reflected or rotated to get new figures.

The original figure is called the **pre-image** and the new figure is called the **image**. Pre-images are denoted by A, B, C … etc., and the images are denoted by A', B', C', … etc. A' can be read as A prime.



Pre-image – A Image – A′

Fig. 4.7

DO YOU KNOW?



Leonardo da Vinci's Vitruvian Man (ca. 1487) is often used as a representation of symmetry in the human body and, by extension, the natural universe.

The operation that maps or moves the pre-image onto the image is called the transformation.

A transformation is a specific set of rules that change the pre-image onto the image.

In this chapter we are going to learn three types of transformation.

4.2.1. Translation:

A *translation* is a transformation that moves all points of a figure in the same distance in the same direction.

Look at the *Fig 4.8*.

From these examples we can observe that all points of a figure move in the same distance and in the same direction.

Using a grid paper, we can specify a translation by how far the shape is moved horizontally and then vertically.

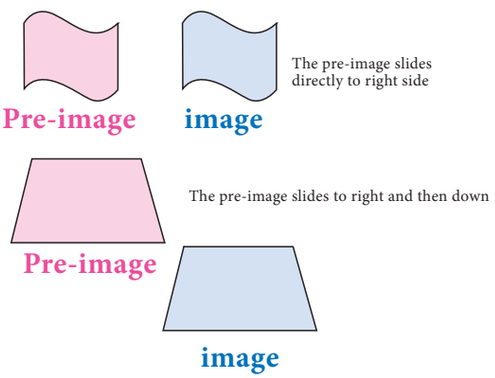


Fig. 4.8

In horizontal, the right side movement is denoted by → and the left side movement is denoted ←.

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